

MID EXAM

Course: Data structures (cs01)

Set: C | Time: 1 Hour | Max Marks: 41 | Date: 14-03-2026

Total Questions: 9 | Answer all questions.

Name: _____

Roll No: _____

Student Sign: _____

Teacher Sign: _____

SECTION A: MCQs

1. Which of the following statements correctly distinguishes between an architectural pattern and a reference model, as defined in the source text? [4 Marks]

- a) An architectural pattern describes element and relation types with constraints, while a reference model outlines functionality.
- b) An architectural pattern is an architecture, whereas a reference model is a standard decomposition of a known problem.
- c) A reference model is derived from an architectural pattern, but an architectural pattern cannot be derived from a reference model.
- d) Architectural patterns primarily focus on mature domains, while reference models are the architect's first major design choice.

2. According to the provided definition and its implications, which of the following statements about software architecture is false? [4 Marks]

- a) Software architecture is an abstraction that suppresses details of elements that do not impact their interaction with other elements.
- b) Every computing system with software possesses a software architecture, even if it is not explicitly known or documented.
- c) The behavior of each element is considered part of the architecture, as it dictates how elements interact.
- d) A system's software architecture is solely defined by a single, dominant structure for clarity and consistency.

3. A software architect is designing a system where real-time interactions, communication protocols, and the potential for deadlocks among executing units are critical concerns. Based on the source text, which view type would be most appropriate for addressing these specific considerations? [4 Marks]

- a) A Module View, specifically a decomposition view.
- b) An Allocation View, particularly a deployment view.
- c) A Component-and-Connector View, focusing on processes and their interactions.
- d) A Quality View, such as a Modifiability view.

4. In a quality attribute scenario, which component identifies the specific part(s) of the system that are affected by a condition and responds to it? [4 Marks]

- a) Source of Stimulus
- b) Stimulus
- c) Environment
- d) Artifact

SECTION A: SHORT ANSWERS

5. Analyze the role of stakeholders in influencing software architectures, detailing at least three specific responsibilities

or actions architects must undertake in relation to them early in the life cycle.

[5 Marks]

6. A development team is documenting a module named Account within their system. Explain three essential properties of a module that should be recorded as part of the supporting documentation for a module view, providing a brief description for each based on the source text.

[5 Marks]

7. Beyond simply redocumenting an architecture, describe three distinct ways the results of architectural reconstruction can be utilized, as outlined in the source text.

[5 Marks]

8. Outline the initial two main steps of the Attribute-Driven Design (ADD) process. For the second step, detail at least three specific substeps involved in refining a module.

[5 Marks]

9. The source text states that "Architecture, by itself, is unable to achieve qualities. It provides the foundation for achieving quality." Explain two reasons why architecture alone cannot achieve qualities and how qualities are ultimately achieved, according to the text.

[5 Marks]

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